

DPT Exercises

Exercise sheet 5

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Exercise 1 (spurious correlation and confounding). A data scientist is investigating the relationship between daily ice cream sales (I) and the number of drowning incidents (D) in a coastal town during the summer. They collect data for 30 days and observe the following joint probabilities for simplified categories (Low/High sales, Low/High incidents):

$P(I = i, D = d) \mid$	$D = \text{Low}$	$D = \text{High}$	$P(I = i)$
$I = \text{Low}$	0.30	0.10	0.40
$I = \text{High}$	0.15	0.45	0.60
$P(D = d)$	0.45	0.55	1.00

To simplify calculations, let's assign numerical values: Low=0, High=1 for both I and D .

- (i) Calculate $\mathbb{E}[I]$, $\text{var}[I]$, $\mathbb{E}[D]$, $\text{var}[D]$, $\text{cov}[I, D]$, and $\rho[I, D]$.
- (ii) Based on $\rho[I, D]$, what do you conclude about the relationship between ice cream sales and drowning incidents?
- (iii) A meteorologist suggests that the average daily temperature (T) is a confounding variable that could be responsible for the observed correlation. Explain conceptually, how a third variable could "create" the observed correlation between I and D . How would you check this and what data would you need? How does that link to the famous saying that "correlation does not imply causation?"

Solution:

- (i) The marginal pmfs are $P(I = 0) = 0.4$, $P(I = 1) = 0.6$ and $P(D = 0) = 0.45$, $P(D = 1) = 0.55$. $\mathbb{E}[I] = 0 \cdot 0.4 + 1 \cdot 0.6 = 0.6$. $\mathbb{E}[D] = 0 \cdot 0.45 + 1 \cdot 0.55 = 0.55$. Since I, D are Bernoulli RVs, $\mathbb{E}[I^2] = \mathbb{E}[I]$ and $\mathbb{E}[D^2] = \mathbb{E}[D]$. $\text{var}[I] = \mathbb{E}[I^2] - (\mathbb{E}[I])^2 = 0.6 - 0.6^2 = 0.24$. $\text{var}[D] = \mathbb{E}[D^2] - (\mathbb{E}[D])^2 = 0.55 - 0.55^2 = 0.2475$. For $\text{cov}[I, D]$, we need $\mathbb{E}[ID] = P(I = 1, D = 1) = 0.45$. $\text{cov}[I, D] = \mathbb{E}[ID] - \mathbb{E}[I]\mathbb{E}[D] = 0.45 - (0.6)(0.55) = 0.45 - 0.33 = 0.12$. $\rho[I, D] = \frac{\text{cov}[I, D]}{\sqrt{\text{var}[I] \text{var}[D]}} = \frac{0.12}{\sqrt{0.24 \cdot 0.2475}} \approx 0.4924$.
- (ii) The correlation $\rho[I, D] \approx 0.4924$ indicates a moderate positive linear association between ice cream sales and drowning incidents. This suggests that days with higher sales tend to also have higher incidents.
- (iii) One could try to check this via the partial correlation coefficients like $\rho[I, D \mid T]$, for which we would need joint probability data for I and D stratified by temperature levels. Conceptually, a third variable T (temperature) could create the observed correlation if T influences both I (e.g., high $T \Rightarrow$ high I) and D (e.g., high $T \Rightarrow$ more swimming $\Rightarrow$ high D) independently. Thus, I and D appear correlated because they both respond to T , not necessarily because I directly causes D or vice-versa. This illustrates that "correlation does not imply causation" because an unobserved common cause (confounder) might be responsible for the statistical association.

Exercise 2 (value at risk (VaR)). An investment analyst models the daily percentage return R of a particular stock as a continuous random variable with pdf

$$p_R(r) = \begin{cases} \frac{2}{15}(r+3) & \text{if } -3 \leq r < 0, \\ \frac{1}{5}(2-r) & \text{if } 0 \leq r \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Since r is in percent, $r = 1$ means a 1% return. The **Value at Risk (VaR)** at confidence level α (or, expressed in percentages, at a $(100\alpha)\%$ confidence level) is the maximum percentage loss L such that the probability of experiencing a loss greater than L is α . Mathematically, if $L > 0$ is the VaR, then $P(R \leq -L) = \alpha$.

- (i) Verify that $p_R(r)$ is a valid pdf.
- (ii) Calculate the cumulative distribution function (CDF), $F_R(r)$. Be sure to define it for all $r \in \mathbb{R}$.
- (iii) Explain how the $(100\alpha)\%$ VaR (expressed as a positive loss percentage) relates to the α -quantile, $r_\alpha = F_R^{-1}(\alpha)$, of the return distribution R .
- (iv) Determine the 5% Value at Risk (VaR) for this investment.
- (v) Calculate the median daily return, the expected daily return, and the daily return that is “most likely,” i.e., the maximum of the pdf. Explain your findings in words.
- (vi) If an investor has \$250,000 in this stock, what is the dollar amount corresponding to the 5% daily VaR you calculated in part (iv)?

Solution:

- (i) The pdf is non-negative (by definition) and integrates as

$$\begin{aligned} \int_{-\infty}^{\infty} p_R(r) dr &= \int_{-3}^0 \frac{2}{15}(r+3) dr + \int_0^2 \frac{1}{5}(2-r) dr \\ &= \frac{2}{15} \left[\frac{r^2}{2} + 3r \right]_{-3}^0 + \frac{1}{5} \left[2r - \frac{r^2}{2} \right]_0^2 \\ &= \frac{3}{5} + \frac{2}{5} = 1. \end{aligned}$$

- (ii) If $r < -3$, $F_R(r) = 0$. If $-3 \leq r < 0$ we have

$$F_R(r) = \int_{-3}^r \frac{2}{15}(t+3) dt = \frac{2}{15} \left[\frac{t^2}{2} + 3t \right]_{-3}^r = \dots = \frac{(r+3)^2}{15}.$$

If $0 \leq r < 2$ we have

$$F_R(r) = F_R(0) + \int_0^r \frac{1}{5}(2-t) dt = \frac{(0+3)^2}{15} + \frac{1}{5} \left[2t - \frac{t^2}{2} \right]_0^r = \dots = \frac{6+4r-r^2}{10}$$

If $r \geq 2$ we have $F_R(r) = F_R(2) = \frac{6+4(2)-2^2}{10} = 1$. So in total

$$F_R(r) = \begin{cases} 0 & \text{if } r < -3 \\ \frac{(r+3)^2}{15} & \text{if } -3 \leq r < 0 \\ \frac{6+4r-r^2}{10} & \text{if } 0 \leq r < 2 \\ 1 & \text{if } r \geq 2 \end{cases}.$$

- (iii) The $(100\alpha)\%$ VaR, expressed as a positive loss L , is such that $P(R \leq -L) = \alpha$. This means $-L$ is the α -quantile of the return distribution, $r_\alpha = F_R^{-1}(\alpha)$. Therefore, $\text{VaR} = -r_\alpha = -F_R^{-1}(\alpha)$.
- (iv) We need to find $r_{0.05}$ such that $F_R(r_{0.05}) = 0.05$. Since $0.05 < F_R(0) = \frac{9}{15} = 0.6$, $r_{0.05}$ must be in $[-3, 0)$. So we have to solve $\frac{(r_{0.05}+3)^2}{15} = 0.05$, which results in $(r_{0.05}+3)^2 = 0.05 \cdot 15 = 0.75$. The two solutions to this are $r_{0.05}^\pm = -3 \pm \sqrt{3}/2$. Since $r_{0.05} \in [-3, 0)$, we eventually find the 5% VaR to be $L = -r_{0.05}^+ = -(-3 + \sqrt{3}/2) \approx 2.134$.

- (v) For the median, we need to solve $F_R(r_{0.5}) = 0.5$, which means $(r_{0.5} + 3)^2/15 = 0.5$ and we find $r_{0.5} = -3 + \sqrt{7.5} \approx -0.2614$.

For the expected value, we compute

$$\begin{aligned}\mathbb{E}[R] &= \int_{-3}^0 r \frac{2}{15}(r+3)dr + \int_0^2 r \frac{1}{5}(2-r)dr \\ &= \frac{2}{15} \left[\frac{r^3}{3} + \frac{3r^2}{2} \right]_{-3}^0 + \frac{1}{5} \left[r^2 - \frac{r^3}{3} \right]_0^2 \\ &= -\frac{9}{15} + \frac{4}{15} = -\frac{1}{3}.\end{aligned}$$

Finally, the most likely return (also called the *mode*) is obtained at the maximum of the pdf, which is at 0%.

While the mode at 0% is the peak of the distribution and thus in a real sense the “most likely” return, the median at -0.2614% says that it is more likely than not to have a loss of at least about a quarter percent. Yet differently, the mean at $-\frac{1}{3}\%$ is pulled even further down towards the longer tail (losses in this case) saying that we should expect on average to make a loss of one third percent. All of these describe different types of “central tendencies” of the distribution.

- (vi) The 5% VaR is approximately 2.134% so the corresponding dollar amount is simply $0.02134 \times \$250\,000 = \$5\,335$. There is a 5% chance of losing \$5 335 or more on a given day with this investment. This is clearly an important metric for investors.

Exercise 3 (lack of memory of the exponential distribution). Let $X : (\Omega, \mathcal{A}, P) \rightarrow (\mathbb{R}_{>0}, \mathcal{B})$ be a random variable such that $X > 0$ with strictly positive density function $p_X(x) > 0$ for all $x \in \mathbb{R}_{>0}$. Show that X is exponentially distributed, i.e., $X \sim \text{Exp}(\lambda)$ for some $\lambda > 0$, if and only if

$$P(X > t + s \mid X > s) = P(X > t) \quad \forall s, t \geq 0. \quad (\text{A})$$

[Hint: Note that the cdf of $\text{Exp}(\lambda)$ is $F_X(x) = 1 - e^{-\lambda x}$ and that $P(X > x) = 1 - P(X \leq x) = 1 - F_X(x)$. Use that the conditional probability can be written as $P(X > t + s)/P(X > s)$. For “ $\Leftarrow$ ” consider $P(X > x) = \int_x^\infty f_X(a) da =: f(x)$ and $\varphi(s) = \frac{f(t+s)}{f(s)}$. How does φ depend on s ?

This states that the exponential function has “no memory” in the sense that an event occurring in the future is independent of how much time has already elapsed: Given that it has not occurred up until time s , the probability of it not occurring in an additional time t is the same as if we started from scratch and waited for time t . The distribution “forgets” how much time has already elapsed. The statement also says that it is the only continuous distribution with that property (on $\mathbb{R}_{>0}$).

Solution:

“ $\Rightarrow$ ”: If $X \sim \text{Exp}(\lambda)$, then $P(X > x) = e^{-\lambda x}$ for $x \geq 0$. Thus, for $s, t \geq 0$:

$$P(X > t + s \mid X > s) = \frac{P(X > t + s \text{ and } X > s)}{P(X > s)} = \frac{P(X > t + s)}{P(X > s)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

“ $\Leftarrow$ ”: Suppose X has a continuous density, such that $P(X > x) = \int_x^\infty p_X(a) da =: f(x)$ is a differentiable function. Note that we assume $f(x) > 0$ for all $x > 0$. For $s \geq 0$ and $t \geq 0$ we define

$$\varphi(s) = \frac{f(t+s)}{f(s)} = \frac{P(X > s+t)}{P(X > s)} \stackrel{(\text{A})}{=} P(X > t) = \frac{P(X > t)}{P(X > 0)} = \frac{f(t)}{f(0)} = \varphi(0).$$

Note what we have done here: We have shown that the function $\varphi(s)$ as defined in the first equal sign is actually constant everywhere and takes the value $\varphi(0)$. Since φ is constant, its derivative vanishes and we obtain via standard differentiation rules

$$0 = \frac{d}{ds} \varphi(s) = \frac{f'(t+s)f(s) - f(t+s)f'(s)}{f(s)^2}.$$

Via some shuffling, we find that

$$\frac{f'(s+t)}{f(s+t)} = \frac{f'(s)}{f(s)} \quad \text{which implies} \quad \frac{f'(x)}{f(x)} = c \in \mathbb{R}.$$

Hence, by integrating both sides (and remembering the derivative of the log ;)), we get

$$\int_0^t \frac{f'(x)}{f(x)} dx = \ln(f(t)) = ct.$$

This implies that $f(t) = \exp(ct)$ and since $\lim_{t \rightarrow \infty} f(t) = \lim_{t \rightarrow \infty} P(X > t) = 0$ (limit of $1 - F_X(t)$), we get $c < 0$ so we can write $c = -\lambda$ for some $\lambda \in \mathbb{R}_{>0}$. Hence, $F_X(x) = 1 - P(X > x) = 1 - \exp(-\lambda t)$ and $p_X(x) = F'_X(x) = \lambda \exp(-\lambda x)$, which means that $X \sim \text{Exp}(\lambda)$.

Exercise 4 (does treatment help? coding + open ended parts). We're aiming to provide exercises that give glimpses of where/how/why probability theory and probabilistic modeling is useful. While highly relevant such exercises would be hard to grade in an exam. The flags "coding" or "open ended parts" indicate that those parts will not be relevant for the final exam.

Note that for Bernoulli variables $X \sim \text{Ber}(p)$, we have that $\mathbb{E}[X] = p(0) \cdot 0 + p(1) \cdot 1 = p(1) = p = P(X = 1)$. So the expected value is also the probability of getting a 1. From finite samples, as in this example, we typically "compute" probabilities empirically just based on the frequencies in the dataset. So when we have n actual binary observations $\{x_i\}_{i=1}^n$ of the RV X with $x_i \in \{0, 1\}$ for each $i \in [n]$, we "compute" (should rather say "estimate") $\mathbb{E}[X]$ simply as the average $\frac{1}{n} \sum_{i=1}^n x_i$. Concretely, in Python using Numpy you could do `np.mean(df['S'])` to get the probability of survival when S is the survival column in a pandas dataframe `df`.

Consider the following settings.

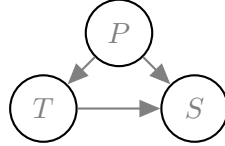
- (i) As data scientist in residence at the university clinic, you are sent a dataset of a clinical study containing three columns with headers *treatment* (T), *survival* (S), *precondition* (P), all containing only binary values. The email says: "Please analyze whether the treatment helped." You can find the data on Moodle called `hospital.csv`. You can use any programming language you like and I somewhat expect you use the LLM/coding assistant/agent of your choice. Do you agree with it?
 - (a) Try to formalize the question in your own words. Take note of what choices and assumptions you have to make in your modeling.
 - (b) Compute the correlation coefficient between treatment and survival. What do you conclude?
 - (c) You are not sure what "precondition" means, but you suspect that people with a precondition may be better off. Compute the correlation coefficient between precondition and survival. What do you conclude?
 - (d) You decide to compute the expected difference in survival probability when getting treatment, compared to no treatment, separately within patients with and without preconditions. Compute these differences and provide a first answer for your task.

(e) The doctors reply: “But overall, among all people, does the treatment help? Are we sure?”
What do you reply?

- (ii) While you draft your reply, the doctors call to let you know they made a mistake and the third column should have been named *complications during treatment* instead of *precondition*. What would you change about your analysis in steps (a) to (e) from (i)? Why?

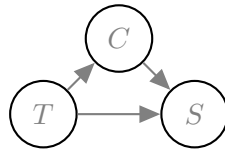
Solution: Please see the python notebook for the solution.

One way to think about this, is to draw what’s called a “causal graph,” a directed graph indicating which variables affect which other variables. A plausible causal graph including the variables *treatment* (T), *survival* (S), *precondition* (P) could be



Somehow, we want to measure whether patients that receive the treatment are more likely to survive than people without, so we want to compare $P(S = 1 \mid T = 1)$ with $P(S = 1 \mid T = 0)$. We can also investigate, whether treatment and survival are positively correlated. In this case, the precondition affects which treatment is chosen (doctors take medical history into account when assigning treatments), but it may also affect the overall chance of survival. So like in the first exercise, the precondition may act as a confounder and “create” or “annihilate” dependence between T and S .

For the new situation, where the precondition becomes a complication (C) a more plausible causal graph is the following.



Here, the complication that occurs during treatment depends on whether treatment is applied and may in turn have an effect on the chance of survival. Here, C is called a “mediator.” Depending on what exactly our question is, in this case the “total effect of T on S ” is captured well by their correlation (which includes effects mediated by C).

Exercise 5 (★ bonus ★). Let $(\Omega, \mathcal{A}, P)$ be a probability space and X, Y, Z real-valued random variables. By means of a counter-example, show the following:

- (i) If X and Y are partially uncorrelated given Z , they are not necessarily independent given Z .
[Hint: Look at Exercise 4.4 and $Z \perp\!\!\!\perp X$.]
- (ii) If X and Y are independent given Z , they are not necessarily partially uncorrelated given Z .
[Hint: Consider $Z \sim \mathcal{U}_{(0,1)}$, $X = Z^2$ and $Y = Z^3$.]

Solution:

- (i) To see this, choose Z independent of $X \sim \mathcal{U}_{(-1,1)}$, $Y = X^2$. Then we obtain

$$\rho[X, Y \mid Z] \propto \underbrace{\rho[X, Y]}_{= \text{cov}[X, Y] = 0} - \underbrace{\rho[X, Z]}_{= 0, \text{as indep.}} \rho[Y, Z] = 0$$

but on the other hand we do not have conditional independence as

$$p(x, y \mid z) \stackrel{X, Y \perp\!\!\!\perp Z}{=} p(x, y) \neq p(x)p(y) = p(x \mid z)p(y \mid z)$$

(ii) Let $Z \sim \mathcal{U}_{(0,1)}$, $X = Z^2$ and $Y = Z^3$. Then: $X \perp\!\!\!\perp Y \mid Z$. This can be seen, using the definition

$$X \perp\!\!\!\perp Y \mid Z \Leftrightarrow F_{X,Y|Z=z}(x, y) = F_{X|Z=z}(x) \cdot F_{Y|Z=z}(y).$$

Conditioning on $Z = z$ ($z \in (0, 1$ fixed but arbitrary) then makes X, Y constant RVs giving

$$F_{X|Z=z}(x) = P(X \leq x \mid Z = z) = \begin{cases} 0 & x < z^2 \\ 1 & x \geq z^2 \end{cases}$$

$$F_{Y|Z=z}(y) = P(Y \leq y \mid Z = z) = \begin{cases} 0 & y < z^3 \\ 1 & y \geq z^3 \end{cases}$$

and also

$$F_{X,Y|Z=z} = P(X \leq x, Y \leq y \mid Z = z) = P(\{X \leq x\} \cap \{Y \leq y\} \mid Z = z) = \begin{cases} 0 & x \leq z^2, y \leq z^3 \\ 0 & x < z^2, y \geq z^3 \\ 0 & x \geq z^2, y < z^3 \\ 1 & x \geq z^2, y \geq z^3 \end{cases}$$

Direct computation yields

$$\sigma[Z] = \sqrt{\mathbb{E}[Z^2] - \mathbb{E}[Z]^2} = \sqrt{\frac{1}{3} - \frac{1}{4}} = \frac{1}{\sqrt{12}}$$

$$\sigma[X] = \sqrt{\mathbb{E}[Z^4] - \mathbb{E}[Z^2]^2} = \sqrt{\frac{1}{5} - \frac{1}{9}} = \frac{2}{\sqrt{45}}$$

$$\sigma[Y] = \sqrt{\mathbb{E}[Z^6] - \mathbb{E}[Z^3]^2} = \sqrt{\frac{1}{7} - \frac{1}{16}} = \frac{3}{\sqrt{112}}$$

$$\text{cov}[X, Z] = \mathbb{E}[Z \cdot Z^2] - \mathbb{E}[Z] \cdot \mathbb{E}[Z^2] = \frac{1}{4} - \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{12}$$

$$\text{cov}[Y, Z] = \mathbb{E}[Z \cdot Z^3] - \mathbb{E}[Z] \cdot \mathbb{E}[Z^3] = \frac{1}{5} - \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{40}$$

$$\text{cov}[X, Y] = \mathbb{E}[Z^2 \cdot Z^3] - \mathbb{E}[Z^2] \cdot \mathbb{E}[Z^3] = \frac{1}{6} - \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

$$\rho[X, Z] = \frac{\sqrt{15}}{4}$$

$$\rho[Y, Z] = \frac{\sqrt{21}}{5}$$

$$\rho[X, Y] = \frac{\sqrt{35}}{6}$$

From this we find $\rho[X, Y] - \rho[X, Z]\rho[Y, Z] = \sqrt{35}/60 \neq 0$.